

Assignment 2:

Stage E - Sensitivity Analysis Report

Name: Rishabh Ray

Student ID: 34525416

Introduction

The Rift Valley trial model is being tested against many constraints, which is helping to minimise the cost of the vaccination trials and helping reduce the time taken to achieve the said objective. All of the models are run in the HiGHS 1.9.0 solver.

All the runs are being run in a 60-second wall time limit.

Table 1: Best Objective Values

Variant	rift1.dzn	rift2.dzn	rift3.dzn	rift4.dzn	rift5.dzn	rift6.dzn	rift7.dzn	rift8.dzn
All Constraints	100	346	168	100	130	184	99	102
Without A1	36	232	168	36	70	168	30	36
Without A2	74	232	168	74	90	168	72	74
Without A3	100	346	168	100	130	184	99	100
Without A4	100	346	168	100	130	184	99	100
Without A5	78	256	168	84	79	168	50	78
Without A6	100	346	168	102	130	186	95	102
Without A7	100	346	168	100	130	168	99	100
Without A8	100	346	168	100	131	184	101	102
Without C1	92	326	134	92	119	164	97	92
Without C2	90	340	146	90	123	178	90	90

This table is written by noting the objective values, running the model against each dataset, removing the constraints one by one from A1-A8 and C1-C2 and checking the values that affect the fluctuations in the objective.

Table 2: Runtime to Prove Optimality (seconds)

Variant	rift1.dzn	rift2.dzn	rift3.dzn	rift4.dzn	rift5.dzn	rift6.dzn	rift7.dzn	rift8.dzn
All Constraints	15s 341msec	>60s	>60s	10s 140msec	>60s	>60s	2s 395msec	12s 883msec
Without A1	4s 795msec	>60s	>60s	5s 408msec	>60	>60s	2s 624msec	4s 467msec
Without A2	6s 595msec	>60s	>60s	3s 950msec	>60s	>60s	8s 633msec	8s 868msec
Without A3	7s 635msec	>60s	>60s	14s 257msec	>60s	>60s	2s 88msec	10s 539msec
Without A4	15s 528msec	>60s	>60s	24s 969msec	>60s	>60s	2s 165msec	19s 337msec
Without A5	1s 589msec	>60s	>60s	4s 147msec	16s 519msec	>60s	6s 475msec	1s 576msec
Without A6	11s 615msec	>60s	>60s	13s 37msec	>60s	>60s	2s 122msec	12s 90msec
Without A7	24s 609msec	>60s	>60s	10s 574msec	>60s	>60s	1s 591msec	7s 185msec
Without A8	18s 262msec	>60s	>60s	13s 9msec	>60s	>60s	4s 834msec	13s 164msec
Without C1	942msec	14s 954msec	35s 3msec	1s 270msec	772msec	28s 993msec	1s 558msec	921msec
Without C2	4s 407msec	>60s	>60s	12s 400msec	35s 741msec	>60s	3s 768msec	4s 454msec

This table contains the time taken to calculate the objective for each dataset when constraints are removed one by one, and how it affects the runtime.

Analysis by Data File

For each dataset, I removed all the constraints from A1-A8 and C1-C2 one at a time and observed that:

rift1.dzn

- Constraint A1 - Removing this constraint, the cost dropped to 36, and it also reduced the runtime. This is likely to happen because removing the vaccine count rule gave the solver more freedom to use fewer vaccines.
- Constraint A2 - This also lowered the cost to 74 and reduced the runtime, probably because the solver was able to reuse the vaccine.s
- Constraint A5 - This significantly reduced the runtime and also the cost by some margin, because the solver was not bounded by the minimum and maximum number of treatments and could make optimal decisions freely

rift2.dzn

- Constraint A1 - Removing the constraint we were able to lower the cost of the objective to 232, but it did not finish in the bounded time, hence optimality was not proven.
- Constraint A2 - Similarly, here the cost was reduced to 232 ,likely due to being able to use the same vaccine for multiple groups and hence less exploration, although the model ran out of the time bound.
- Constraint C1 - It was the only constraint upon removal of which we were able to provide the optimal solution. This is because we did not have to calculate different costs for all the groups.

rift3.dzn

- Constraint C1: Removing the requirement for all group expenses to be different the solver can produce a result quickly by taking on values for each group, which can be same.
- Constraint C2: Since the solver can take more values in group expenses, which do not have to be within any bounds, the cost decreased somewhat.
- Other constraints did not have any major impact on the model objective.

rift4.dzn

- Constraint A1: A great improvement was brought about in the objective cost by removing the scheduling of a vaccine, which helps the solver get a much lower objective cost.
- Constraint A5: The cost decreased from 100 to 79. Most likely, the model avoided costly required treatments.
- Constraint C2: By allowing one group to pay a larger portion of the costs, the cost was lowered to 90 by eliminating the cost of being close to each group rule.

rift5.dzn

- Constraint A1 - Removing the A1 constraint brought down the cost to 70, which confirmed that having a schedule for the vaccines made a difference to the objective of the model.
- Constraint A5 - Removing this constraint had an impact where the cost was reduced to 79. - This indicates that there was a strong impact from treatment quotas on scheduling costs.
- Constraint C1 - Removing this help solve the model in much less runtime, by providing a group cost that was not different from each other.

rift6.dzn

- Constraint C1: By removing this restriction, the runtime of 28.993s was achieved to find the optimal solution. Therefore signifying that keeping the cost of all groups different had an effect on the total cost.
- Constraint A1,A2,A5: By removing these constraints one by one we were able to achieve a lower objective cost although they all made a minor difference to the model.

rift7.dzn

- Constraint A1: It shows the most impact on the objective cost where the cost is brought down to 30, signifying the higher calculation power used to calculate the 0-2 vaccination schedule.
- Constraint A5: The price decreased to 50 on removing the A5 constraint. Signifying that the plans that underwent less testing were more effective.
- Constraint C2: By eliminating the budget difference between groups, there is a slight cost and a great runtime benefit.

rift8.dzn

- Constraint A1: The Cost decreased to 36 when the treatment bounds on the vaccinations were removed.
- Constraint A2: Removal of this constraint contributed to a further cost reduction to 74, signifying that removing the different variations of vaccine in groups is providing good objectives.
- Constraint C1: On removal of this constraint, the runtime drastically reduced to 921ms, showing a great impact on the removal of the different costs for each group

5. Overall Constraint Importance

Based on the constraint-wise removal analysis across all datasets, the following ranking of the constraints reflects both the impact on cost reduction and solver performance:

Rank	Constraint	Why it matters
1	A5 (min/max treatment frequency)	By regularly eliminating this, the solver was able to eliminate costly test sequences, significantly increasing flexibility..
2	A1 (duplication rule)	Costs were significantly reduced by eliminating the requirement for exactly two uses per vaccine, which added major scheduling complication.
3	C1 (unique group costs)	Frequently enables faster runtime when disabled, demonstrating how search interferes by imposing a unique cost.
4	C2 (cost proximity)	Prevented cost inequalities between groups; when group testing constraint is changed, its removal produced better results.

Other constraints such as A2, A4, A6–A8 influenced structure or sequencing but were less dominant across files.

Conclusion

This sensitivity analysis showed me that a few constraints have a significant impact on solver performance and solution quality, especially **A5 (treatment frequency limits)**, **A1 (vaccine duplication)**, and **C1 (unique group costs)**. By allowing the solver to get around strict test scheduling, removing A5 resulted in the biggest cost savings across practically all datasets. Significant effects were also produced by A1 and C1, either by reducing budget differences between groups or by simplifying symmetry.

The effects of other constraints, such as A2, A6, and C2, were either dataset-specific or very subtle.